Tutorial-1

A smart farming system predicts whether a plant will grow well (1) or poorly (0) based on daily sunlight hours and water amount in liters.

Dataset:

Sunlight Hours (x_1)	Water Liters (x_2)	Bias (x_3)	Target (Good Growth = 1 / poor = 0) (y)
3	1	1	0
8	4	1	1
5	2	1	0
10	5	1	1
7	3	1	1

① Understand and Explain the data

Inputs (x_1 & x_2) Output (Target)

mean of $x_1 = \mu x_1 = 6.6$

mean of $x_2 = \mu x_2 = 3$

* Variance; $\boxed{\sigma^2 = \frac{1}{N} \Sigma (x_i - \mu)^2}$

where N = No. of Data

x_i = Input

μ = mean

$$\frac{(x_i - \mu x_i)^2 + (x_i - \mu x_i)^2 + \dots}{\text{no. of } x \text{ values}}$$

$$= \frac{(x_{11} - \mu x_1)^2 + (x_{12} - \mu x_1)^2 + (x_{13} - \mu x_1)^2 + (x_{14} - \mu x_1)^2 + (x_{15} - \mu x_1)^2}{5}$$

$$= \frac{(3-6.6)^2 + (8-6.6)^2 + (5-6.6)^2 + (10-6.6)^2 + (7-6.6)^2}{5}$$

$$= 5.84$$

$$\Rightarrow \frac{(x_i - \mu_i)^2 + (x_i - \mu x_i)^2 + \cdots}{\text{no. of values}}$$

$$= \frac{(x_{21} - \mu x_2)^2 + (x_{22} - \mu x_2)^2 + (x_{23} - \mu x_2)^2 + (x_{24} - \mu x_2)^2 + (x_{25} - \mu x_2)^2}{5}$$

$$= \frac{(1-3)^2 + (4-3)^2 + (2-3)^2 + (5-3)^2 + (3-3)^2}{5}$$

$$= \frac{(-2)^2 + (1)^2 + (-1)^2 + (2)^2 + (0)^2}{5}$$

$$= \frac{4+1+1+4+0}{5} = \frac{10}{5} = 2.$$

variance of $x_1 \Rightarrow \sigma^2 x_1 = 5.84$.

variance of $x_2 \Rightarrow \sigma^2 x_2 = 2$

$$\sigma^2 x_1 > \sigma^2 x_2$$

Standard deviation of.

$$\sqrt{\sigma x_i^2} = \sigma x_1 = 2.40$$

Standard deviation of.

$$\sqrt{\sigma x_2^2} = \sigma x_2 \Rightarrow 1.414.$$

The factor whose variance is higher is important

$\therefore$ The Sunlight Hours > water Liters

ML Model

perceptron Model:

It is a simple Nural Network model

It is a simple decision making model that can Ansuer yes/No

perceptron:

① Inputs : x_1 & x_2

② weights associated with input : w_1 & w_2

③ Bias -> It gives the model flexibility (b)
almost it is 1

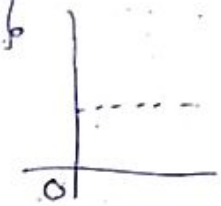
④ weighted sum : $z = w_1 x_1 + w_2 x_2 + b = 1$

$= \sum_i w_i x_i + b$ $\qquad w_1$ & $w_2 = 0$

⑤ Activation function -> It is an mathematical function followed by ML
(step function)

step function is a easiest function with (0 & is)

Activation function: $y^{\wedge} = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{if } z < 0 \end{cases}$



⑥ Updated weight :

$w_i^{new} = w_i^{old} + \eta (y - y^{\wedge}) x_i$

eta $\quad 0.5$

$z = 0(3) + 0(1) + 1$

$z = 1 < 0 \quad 0$

$\geq 0 \quad 1$

Y = Actual Target / output

$y^{\wedge}$ = predicted target / output

$\boxed{\eta = \text{Learning Rate}}$

$w_1 = 0 - 1(0-1)3$

$= 0 + 3$

$w_1 = -3$

$w_2 = 0 - 1(0-1) \cdot 1$

$w_2 = -1$

$z = (-3)(3) + (-1)(0) + 1$

$= -9 - 0 + 1$

$= -8 < 0 \quad 0$

prediction:

① Initial

Let ; $w_1 = 0$, $w_2 = 0$, $\eta = 1$ we can take 0 - a also.

Inputs = $x_1 = 3$, $x_2 = 1$, $b = 1$

$z = ?$

$z = w_1 x_1 + w_2 x_2 + b$

$(0)(3) + (0)(1) + 1 = 1$

$z = 1$, so check is $y^n = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{if } z < 0 \end{cases}$

$y^n = 1$

but the target is "a" so we want to change the weights

$W_1^{new} = 0 + 1(0-1) \cdot 3 = -3$

$w_2^{new} = 0 + 1(0-1) \cdot 1 = -1$

2nd case for 2nd sunlight & water inputs

Inputs: $x_1 = 8$, $x_2 = 4$, $b = 1$, $w_1 = -3$, $w_2 = -1$, $\eta = 1$, $b = 1$

$z = (-3)(8) + (-1)(4) + 1$

$= -24 - 4 + 1$

$= -28 + 1$

$= -27$

$\boxed{z = -27}$; check y^n $\boxed{y^n = 0}$

But in Dataset of 2nd row the target is 1 so it do not matched

$w_1^{new} = (-3) + 1(1 - 0)8$

$= -3 + (1-0)8$

$= -3 + 8$

$w_1^{new} = 5$

$w_2 new = -1 + 1(1-0)(4)$

$w_2 New = -1 + 4$

$w_2 New = 3$

$x_1 = 5$ $x_2 = 2$ $b = 1$ $w_1 = 5$ $w_2 = 3$ $\eta = 1$

$z = w_1 x_1 + w_2 x_2 + b$

$(5)(5) + (3)(2) + 1$

$25 + 6 + 1$

$z = 32 \quad z > 0 \Rightarrow y^n = 1$

$w_1 new = (5) + 1(0-1)(5)$

$w_1 new = 5 + (-5)$

$\boxed{w_1 new = 0}$

$w_2 new = (3) + 1(0-1)(2)$

$w_2 new = 3 - 2$

$\boxed{w_2 new = 1}$

$x_1 = 10, \; x_2 = 5, \; b = 1, \; w_1 = 0, \; w_2 = 1, \; \eta = 1$

$z = w_1 x_1 + w_2 x_2 + b$

$= (0)(10) + (1)(5) + 1$

$= 0 + 5 + 1$

$\boxed{z = 6} \quad z > 0$ so $\boxed{y^n = 1}$

$\therefore$ $\boxed{\text{activation function} = \text{old target}}$